第八章
理想气体的热力过程

Thermodynamic Process of Ideal-Gas



作业

教材习题：

8-5、8-7、8-10、8-11、8-14

补充习题：

8-12

第八章 理想气体的热力过程

§8-1 概述

§8-2 四种典型热力过程的分析

§8-3 多变过程

§8-4 热力过程的热二律分析举例

§8-1 概述

1. 对象

理想气体　可逆过程　定比热　1kg工质

2. 目的　提高热力学过程的热功转换效率

确定热力过程中工质状态变化的规律

过程中的能量交换

3. 方法

基本过程

- 抽象分类 Ⓟ Ⓥ Ⓣ Ⓢ　ⓝ
- 可逆过程 (不可逆再修正)

§8-1 概述

4. 内容

1）确定过程方程式 —— 过程中参数变化关系

参数 (p, T, v) 变化关系

2）计算$\triangle u$, $\triangle h$, $\triangle s$

3）计算能量转换关系q, w（包括w_v, w_t）

4）在p-v及T-s 图上表示

研究热力学过程的依据

1) 热一律 $\delta q = du + \delta w_v = dh + \delta w_t$

稳流 $q = \Delta h + \frac{1}{2}\Delta c^2 + g\Delta z + w_s$

2) 理想气体 $pv = RT \quad c_p - c_v = R \quad k = \frac{c_p}{c_v}$

$u = f(T) \quad h = f(T)$

3) 可逆过程 $w_v = \int p dv \quad w_t = -\int v dp$

$q = \int T ds$

§8-2 四种典型热力过程的分析

1. 定容过程

(1) 过程方程

$$v = const$$

$$pv = RT \quad \Rightarrow \quad \frac{p}{T} = const$$

(2) 基本状态参数关系

$$v_1 = v_2$$

$$\frac{p_2}{p_1} = \frac{T_2}{T_1}$$

§8-2 四种典型热力过程的分析

1. 定容过程

(3) 状态参数变化

$$\Delta u = \int_{T_1}^{T_2} c_v dT = c_v (T_2 - T_1)$$

$$\Delta h = c_p (T_2 - T_1) = (c_v + R)(T_2 - T_1)$$

$$\Delta s = c_v \ln \frac{T_2}{T_1}$$

§8-2 四种典型热力过程的分析

1. 定容过程

(4) 功量及热量

容积功 $w_v = \int_1^2 p\,dv = 0$

技术功 $w_t = -\int_1^2 v\,dp = v(p_1 - p_2) = R(T_1 - T_2)$

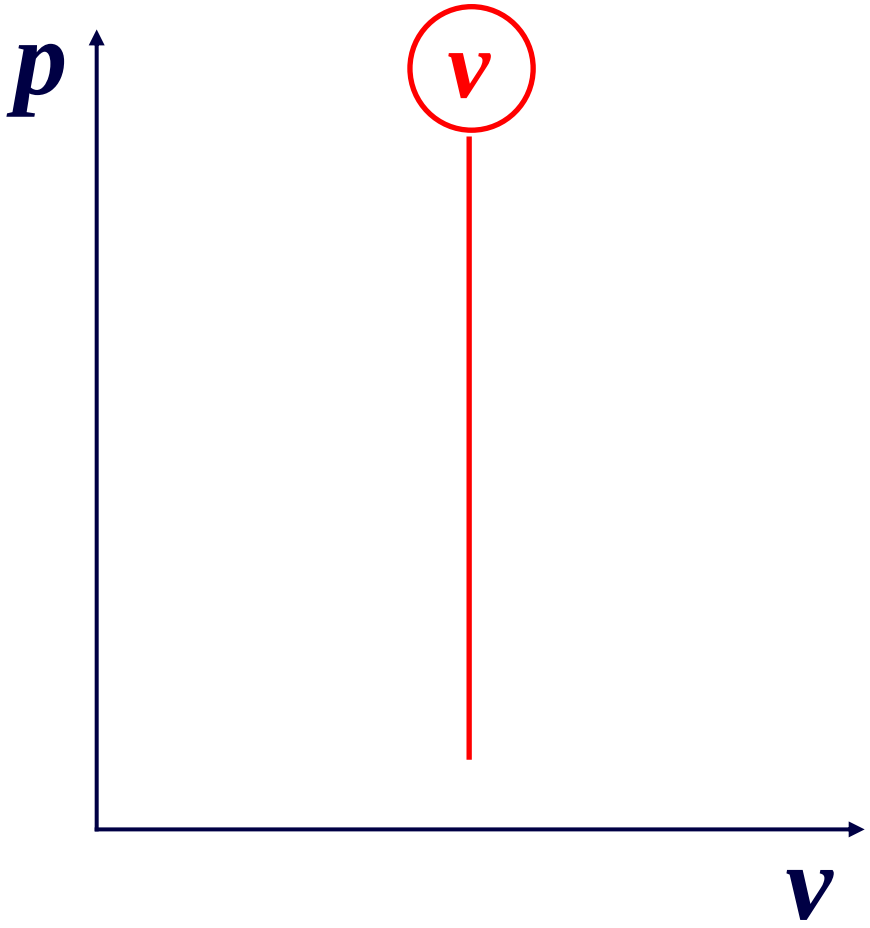
热量 $q = \Delta u = c_v(T_2 - T_1)$

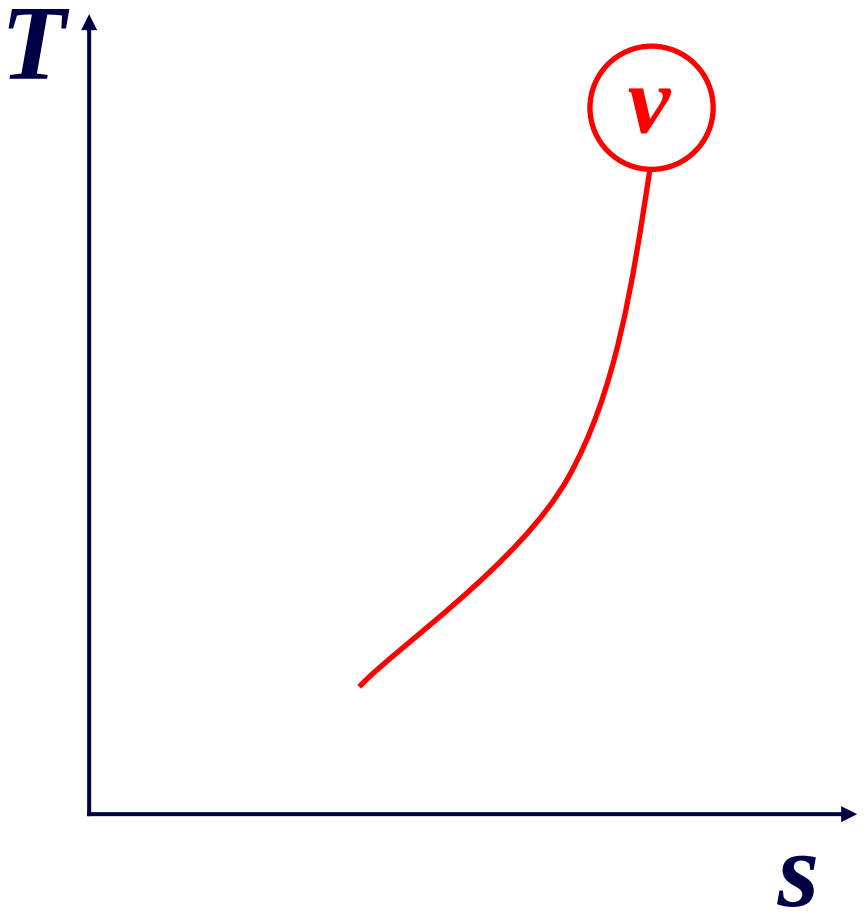
(5)状态图

$$\left(\frac{\mathrm{d}T}{\mathrm{d}s}\right)_v = \frac{T}{c_v}$$

$$T\mathrm{d}s = c_v\mathrm{d}T + p\mathrm{d}v$$

上凸or上凹？ T↑ 斜率↑





§8-2 四种典型热力过程的分析

2. 定压过程

(1) 过程方程

$$p = const$$

$$pv = RT \quad \Rightarrow \quad \frac{v}{T} = const$$

(2) 基本状态参数关系

$$p_1 = p_2$$

$$\frac{v_2}{v_1} = \frac{T_2}{T_1}$$

§8-2 四种典型热力过程的分析

2. 定压过程

(3) 状态参数变化

$$\Delta u = c_v(T_2 - T_1) = (c_p - R)(T_2 - T_1)$$

$$\Delta h = c_p(T_2 - T_1)$$

$$\Delta s = c_p \ln \frac{T_2}{T_1}$$

§8-2 四种典型热力过程的分析

2. 定压过程

(4) 功量及热量

容积功 $w_v = \int_1^2 p\,dv = p(v_2 - v_1) = R(T_2 - T_1)$

技术功 $w_t = -\int_1^2 v\,dp = 0$

热量 $q = \Delta h = c_p(T_2 - T_1)$

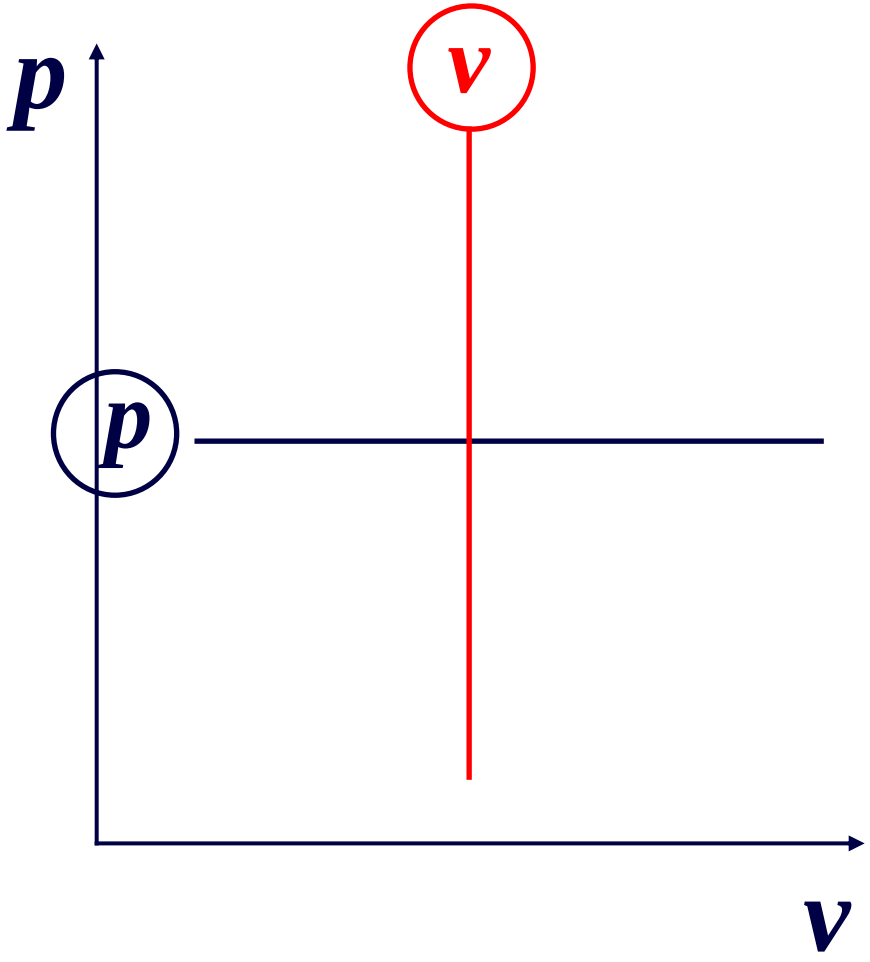
$= \Delta u + p(v_2 - v_1)$

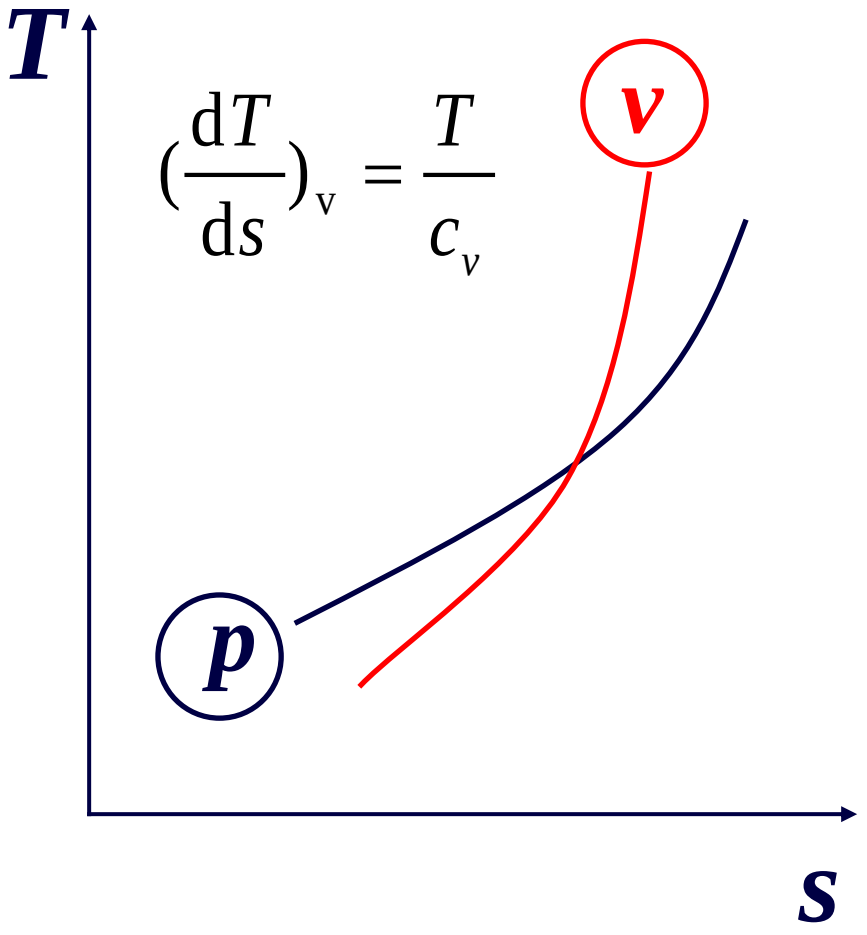
(5) 状态图

$$\left(\frac{\mathrm{d}T}{\mathrm{d}s}\right)_p = \frac{T}{c_p}$$

$$T\mathrm{d}s = c_p\mathrm{d}T - v\mathrm{d}p \qquad c_p > c_v$$

上凸or上凹？ T ↑ 斜率 ↑





§8-2 四种典型热力过程的分析

3. 定温过程

(1) 过程方程

$$T = const$$

$$pv = RT \quad \Rightarrow \quad pv = const$$

(2) 基本状态参数关系

$$T_1 = T_2$$

$$\frac{p_1}{p_2} = \frac{v_2}{v_1}$$

§8-2 四种典型热力过程的分析

3. 定温过程

(3) 状态参数变化

$$\Delta u = 0$$

$$\Delta h = 0$$

$$\Delta s = R \ln \frac{v_2}{v_1} = R \ln \frac{p_1}{p_2}$$

§8-2 四种典型热力过程的分析

3. 定温过程

(4) 功量及热量

$$w_v = w_t = q$$

容积功 $$w_v = \int_1^2 p\mathrm{d}v = \int_1^2 \frac{RT}{v}\mathrm{d}v = RT\ln\frac{v_2}{v_1}$$

技术功 $$w_t = -\int_1^2 v\mathrm{d}p = -\int_1^2 \frac{RT}{p}\mathrm{d}p = RT\ln\frac{p_1}{p_2}$$

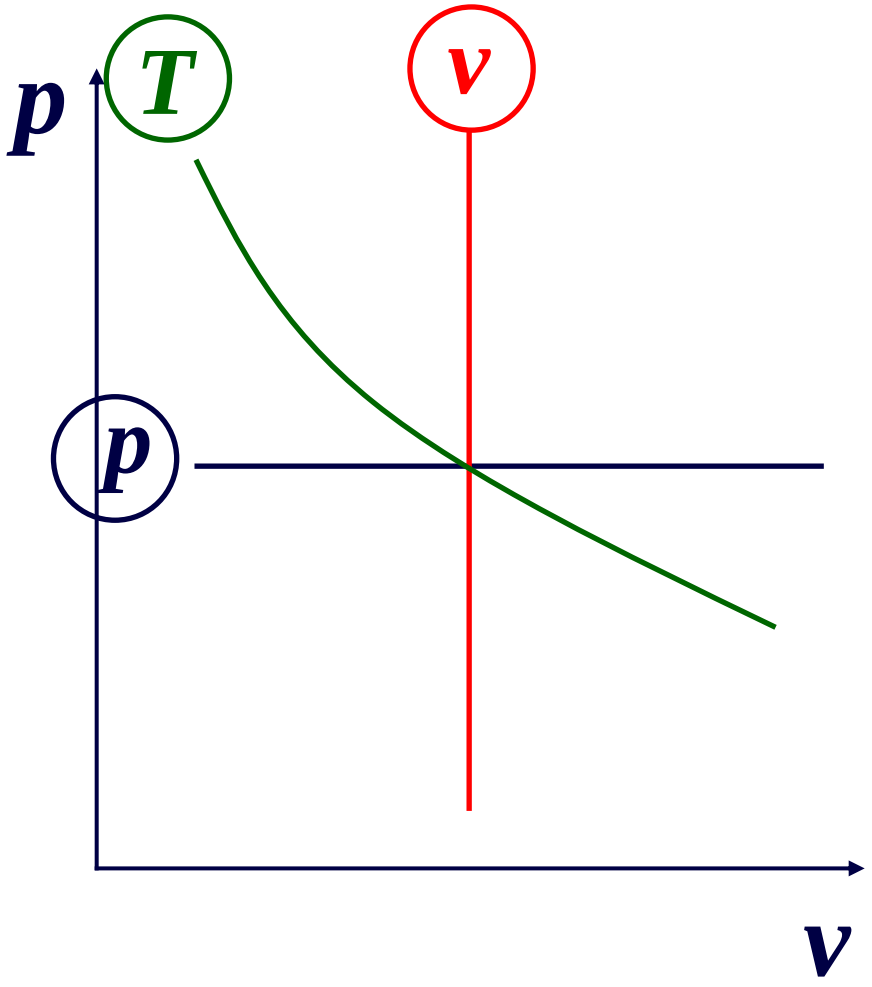
热量 $$q = \int_1^2 T\mathrm{d}s = T(s_2 - s_1)$$

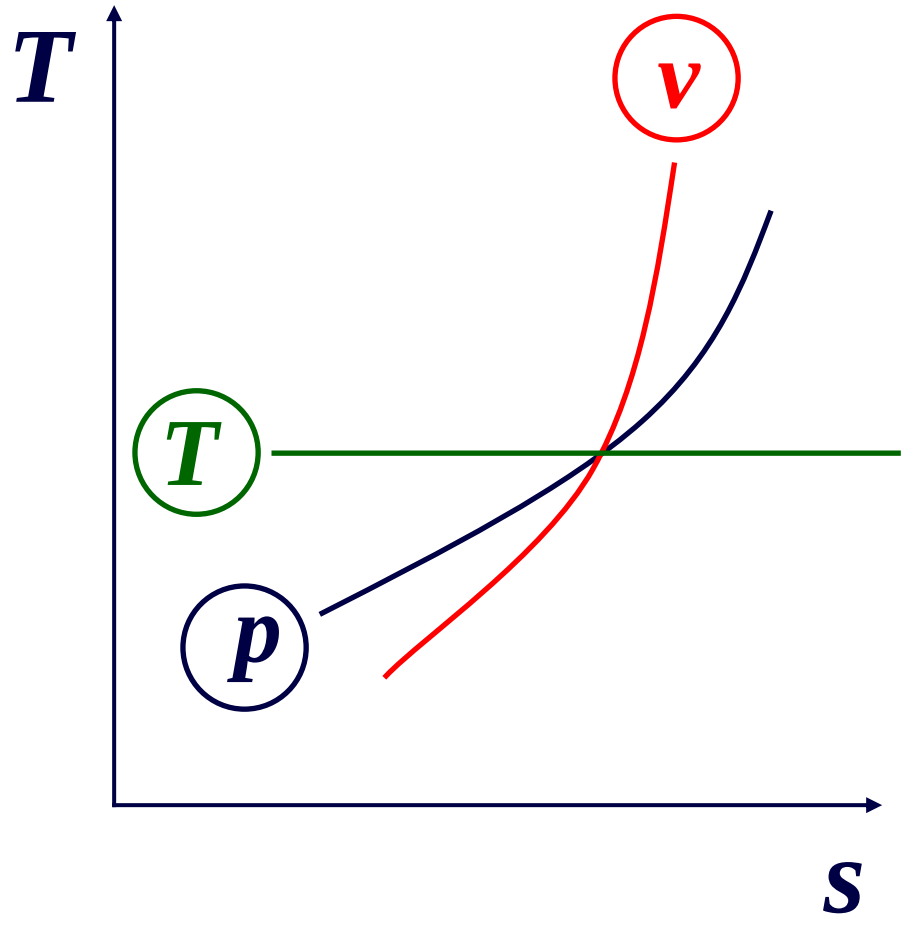
(5)状态图

$$\left(\frac{\mathrm{d}p}{\mathrm{d}v}\right)_T = -\frac{p}{v}$$

$pv = c \quad p\mathrm{d}v + v\mathrm{d}p = 0$

上凸or上凹？ p↑ 斜率↑





§8-2 四种典型热力过程的分析

4. 绝热过程

可逆 $ds = \dfrac{\delta q_R}{T}$ 绝热

$ds = 0$ (S)

不能说绝热过程就是等熵过程, 必须是可逆绝热过程才是等熵过程。

(1)过程方程

$ds = 0$

理想气体

$$ds = c_v \frac{dp}{p} + c_p \frac{dv}{v} = 0$$

$$k = \frac{c_p}{c_v}$$

$$\frac{dp}{p} + k\frac{dv}{v} = 0$$

当 $k = const$

$$\ln p + k \ln v = const$$

$$\boxed{pv^k = const}$$

三个条件：

(1)理想气体 (2)可逆过程 (3)k为常数

(1)过程方程

$$pv^k = const$$

$$pv^k = (pv)v^{k-1} = RTv^{k-1} = const$$

$$Tv^{k-1} = const$$

$$pv^k = \frac{p^k v^k}{p^{k-1}} = \frac{(RT)^k}{p^{k-1}} = const$$

$$Tp^{\frac{1-k}{k}} = const$$

(2) 状态参数关系

$$pv^k = const \Rightarrow \frac{p_2}{p_1} = \left(\frac{v_1}{v_2}\right)^k$$

$$Tv^{k-1} = const \Rightarrow \frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{k-1}$$

$$\frac{T}{p^{\frac{k-1}{k}}} = const \Rightarrow \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}$$

(3) 状态参数变化

状态参数的变化与过程无关

内能变化 $\Delta u = \int c_v dT = c_v (T_2 - T_1)$

焓变化 $\Delta h = \int c_p dT = c_p (T_2 - T_1)$

熵变化 $\Delta s = 0$

(4) 功量及热量

膨胀功 $pv^k = c$

$$w_v = \int p\,dv = \int \frac{c}{v^k}dv = \frac{c}{1-k}v^{1-k}\Big|_1^2 = \frac{1}{1-k}(p_2v_2 - p_1v_1)$$

$$= \frac{R}{k-1}(T_1 - T_2) = c_v(T_1 - T_2) = -\Delta u \qquad q = \Delta u + w$$

技术功

$$w_t = -\int v\,dp = -\Delta h = c_p(T_1 - T_2) = kw_v \qquad q = \Delta h + w_t$$

热量 $q = 0$

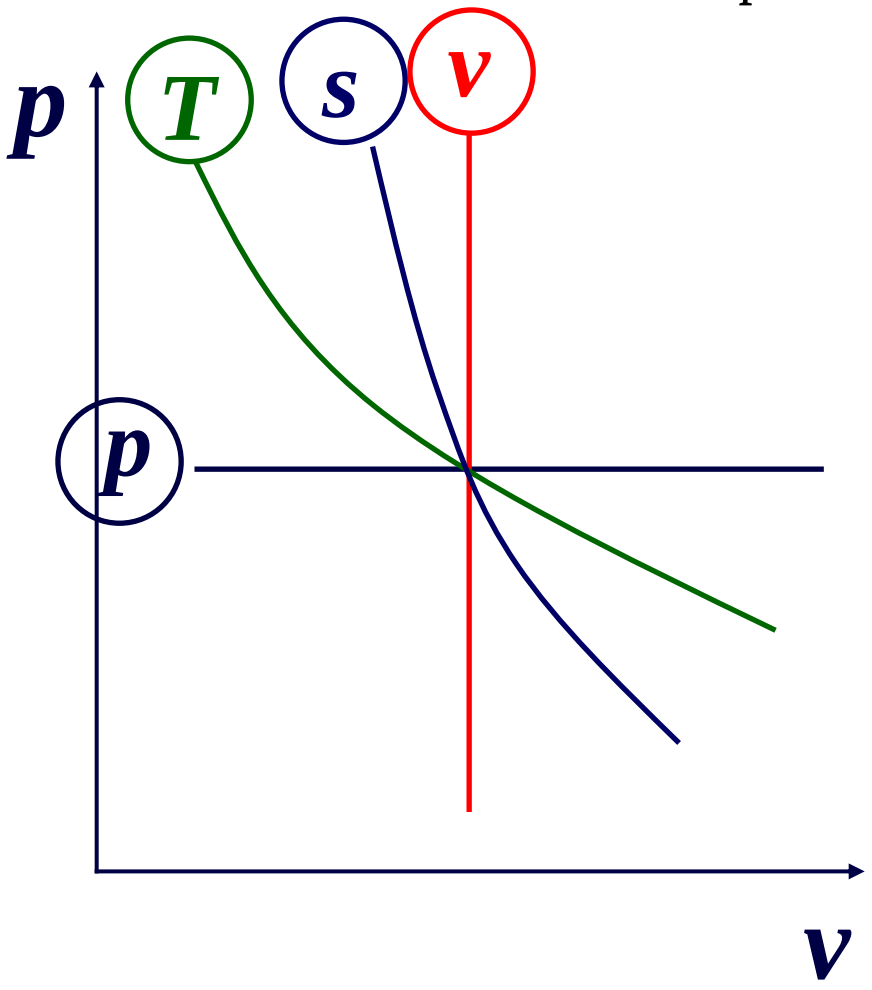
(5) 状态图

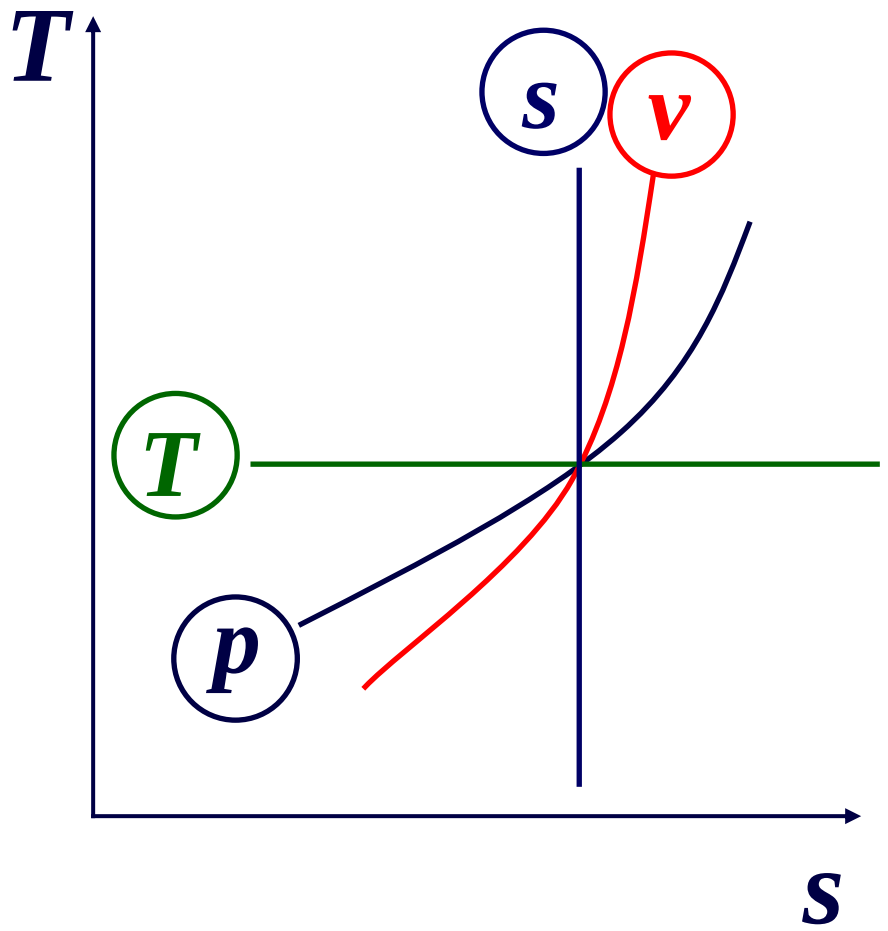
$$\left(\frac{\mathrm{d}p}{\mathrm{d}v}\right)_s = -\frac{kp}{v}$$

$$pv^k = c \quad \mathrm{d}(pv^k) = 0$$

$$kpv^{k-1}\mathrm{d}v + v^k\mathrm{d}p = 0$$

$$\left(\frac{\mathrm{d}p}{\mathrm{d}v}\right)_T = -\frac{p}{v}$$





§8-3 多变过程 Polytropic process

多变过程是一些有规律的过程的总称

1. 过程方程

$$pv^n = const$$

*n*是常量，
每一过程有一 *n* 值

n

$$Tv^{n-1} = const \qquad Tp^{\frac{1-n}{n}} = const$$

2. 基本状态参数关系

$$\frac{p_2}{p_1} = \left(\frac{v_1}{v_2}\right)^n \qquad \frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{n-1} \qquad \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$

3. 状态参数变化

状态参数的变化与过程无关

内能变化 $\Delta u = c_v(T_2 - T_1)$

焓变化 $\Delta h = c_p(T_2 - T_1)$

熵变化

$$\Delta s = c_v ln\frac{T_2}{T_1} + Rln\frac{v_2}{v_1} = c_p ln\frac{T_2}{T_1} - Rln\frac{p_2}{p_1}$$

$$= c_p ln\frac{v_2}{v_1} + c_v ln\frac{p_2}{p_1}$$

4. 功量及热量

$$w_v = \int p dv = \frac{R}{n-1}(T_1 - T_2) \qquad p v^n = const$$

$$w_t = n w_v$$

$$q = \Delta u + w_v = c_v(T_2 - T_1) - \frac{R}{n-1}(T_2 - T_1)$$

$$= (c_v - \frac{R}{n-1})(T_2 - T_1) = \frac{n-k}{n-1} c_v (T_2 - T_1) = c_n (T_2 - T_1)$$

$$c_n = \frac{n-k}{n-1} c_v$$

多变过程比热容

多变过程与基本过程的关系

n **p** **T** **s** **v**

$pv^n = const$ $\quad c_n = \dfrac{n-k}{n-1}c_v$ $\quad c_n = \dfrac{1-\frac{k}{n}}{1-\frac{1}{n}}c_v$

(1) 当 n = 0 $pv^0 = const \Rightarrow p = C$ $\quad c_n = kc_v = c_p$ **p**

(2) 当 n = 1 $pv^1 = const \Rightarrow T = C$ $\quad c_n = \infty$ **T**

$pv = RT$

(3) 当 n = k $pv^k = const \Rightarrow s = C$ $\quad c_n = 0$ **s**

(4) 当 n = ∝ $p^{\frac{1}{n}}v = const \Rightarrow v = C$ $\quad c_n = c_v$ **v**

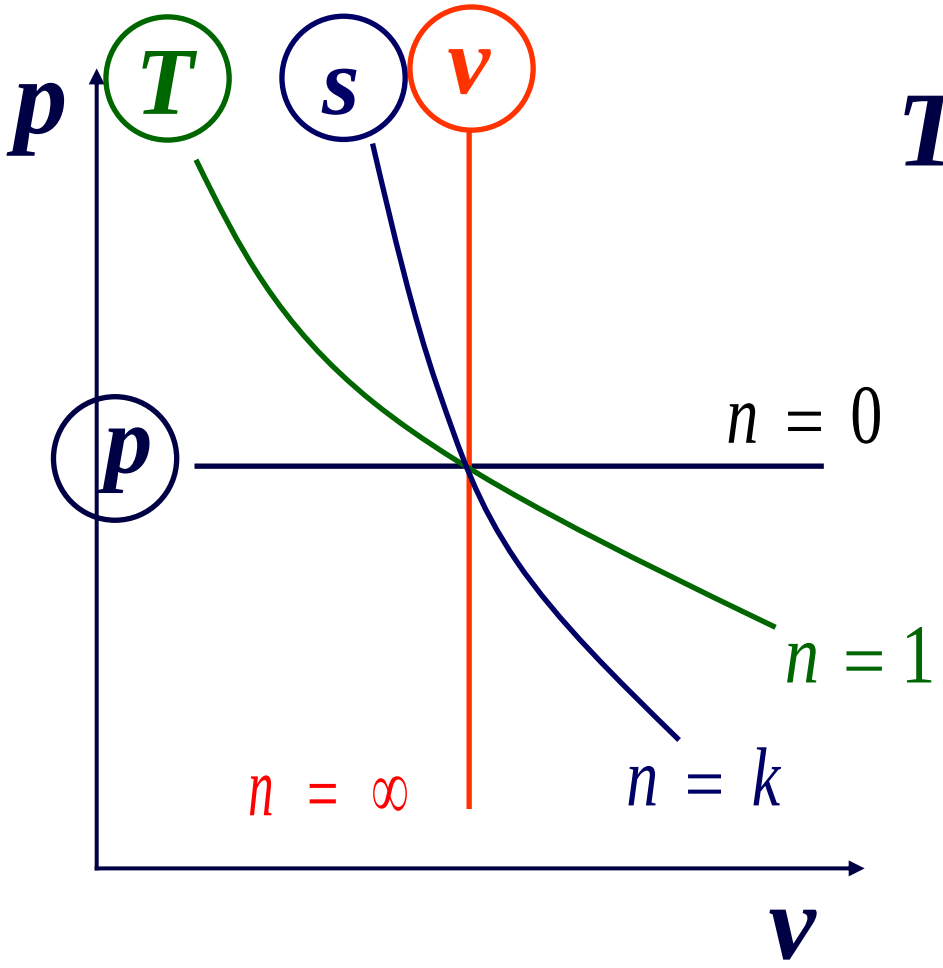
基本过程是多变过程的特例

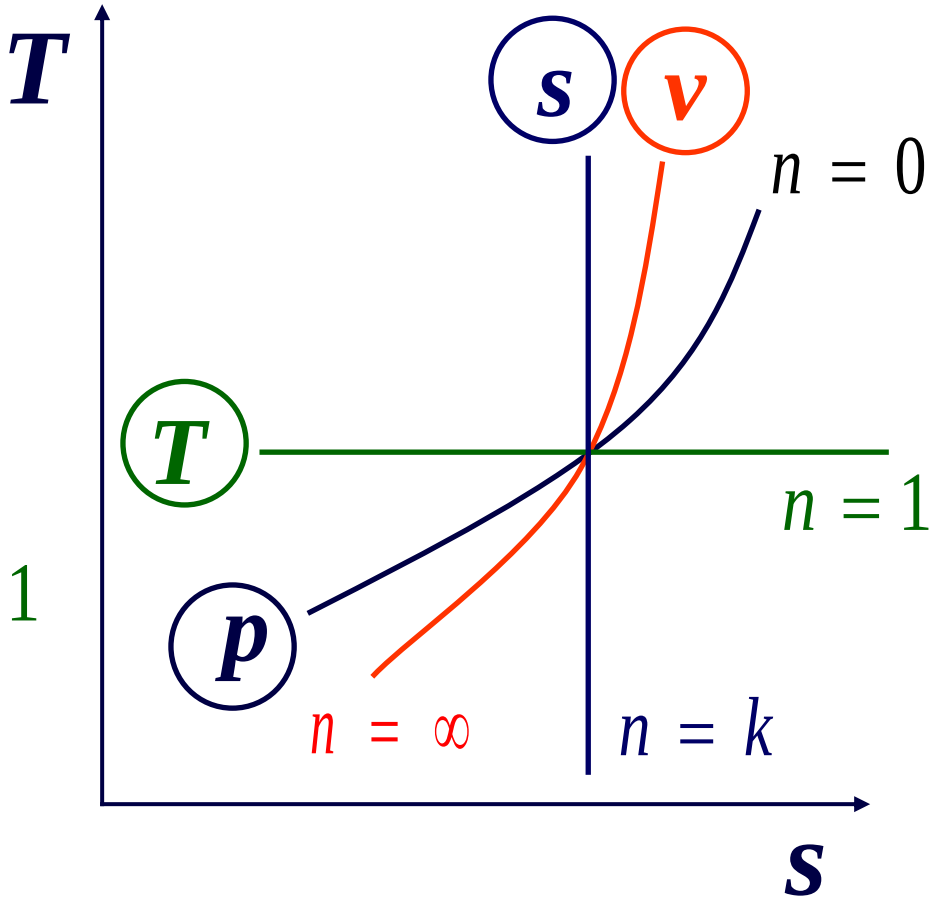
理想气体基本过程的p-v,T-s图

$$\frac{dp}{dv}=-\frac{np}{v}$$

$$\frac{dT}{ds}=\frac{T}{c_n}$$

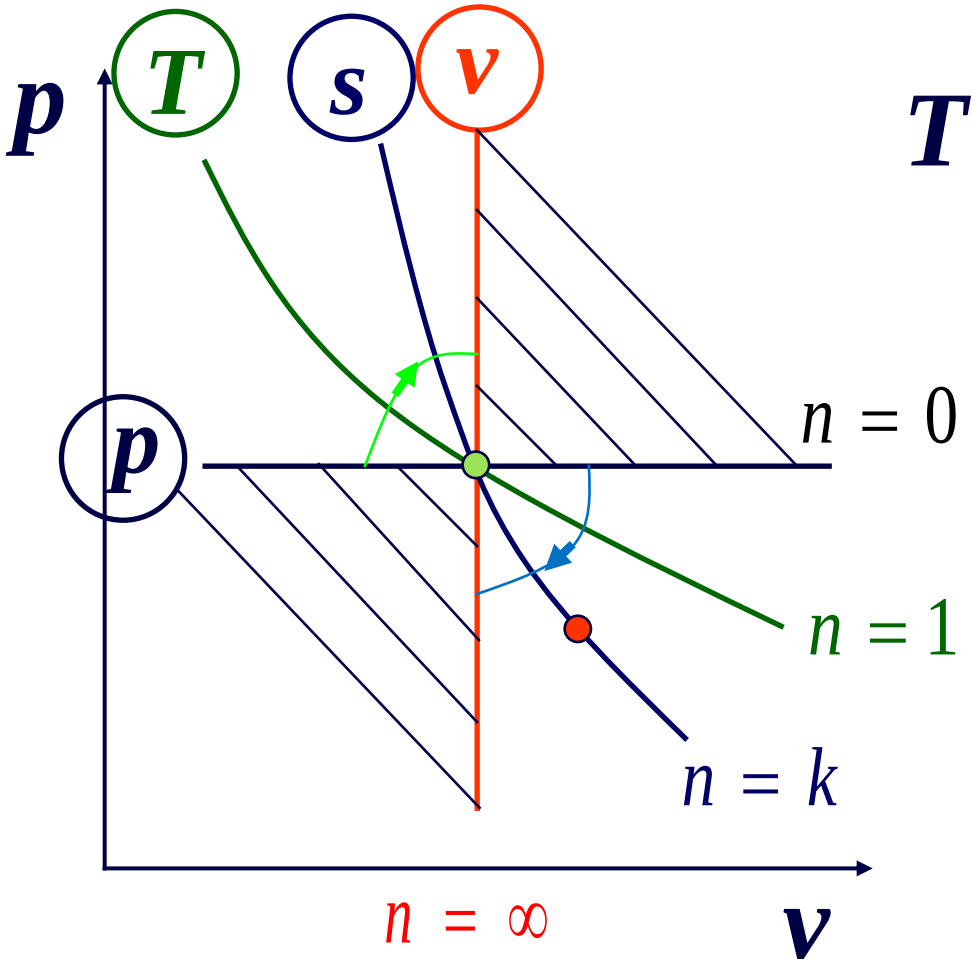
$$c_n=\frac{n-k}{n-1}c_v$$

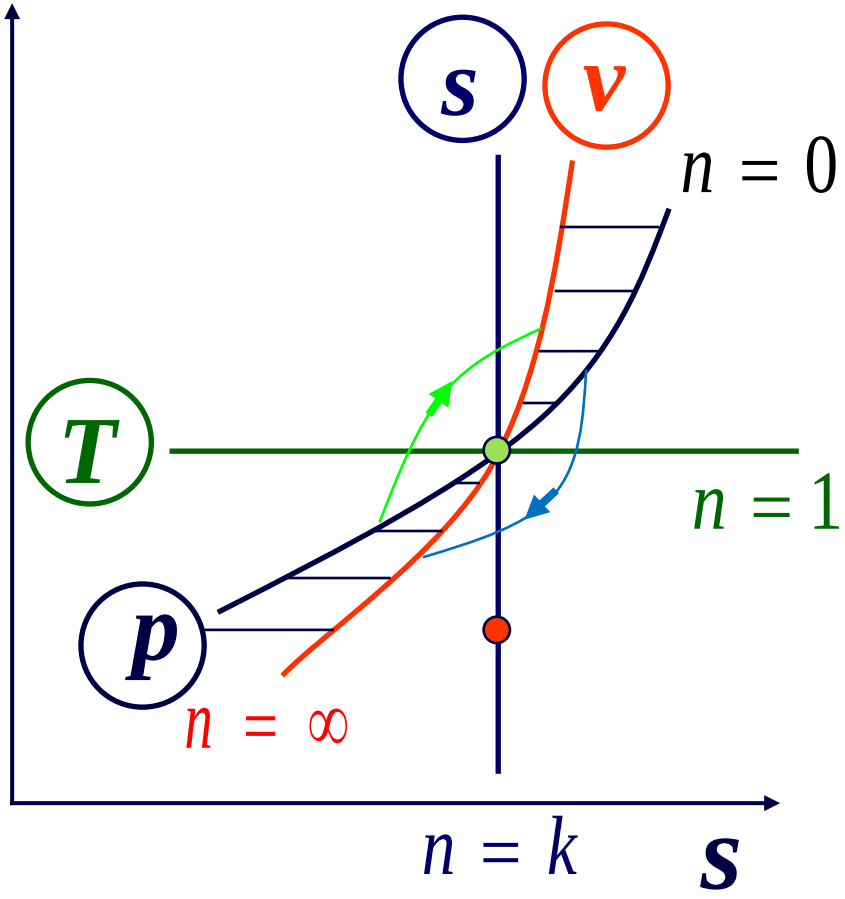




理想气体基本过程的p-v,T-s图

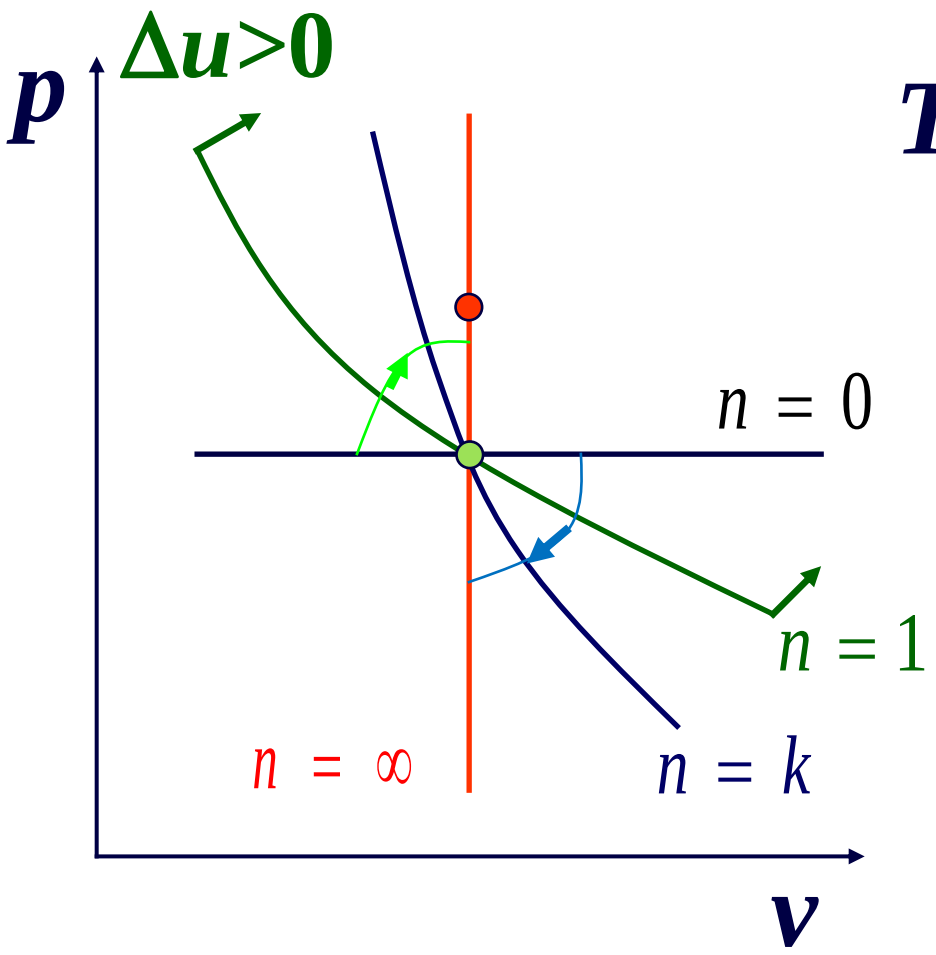
$$\frac{T_2}{T_1}=\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}$$

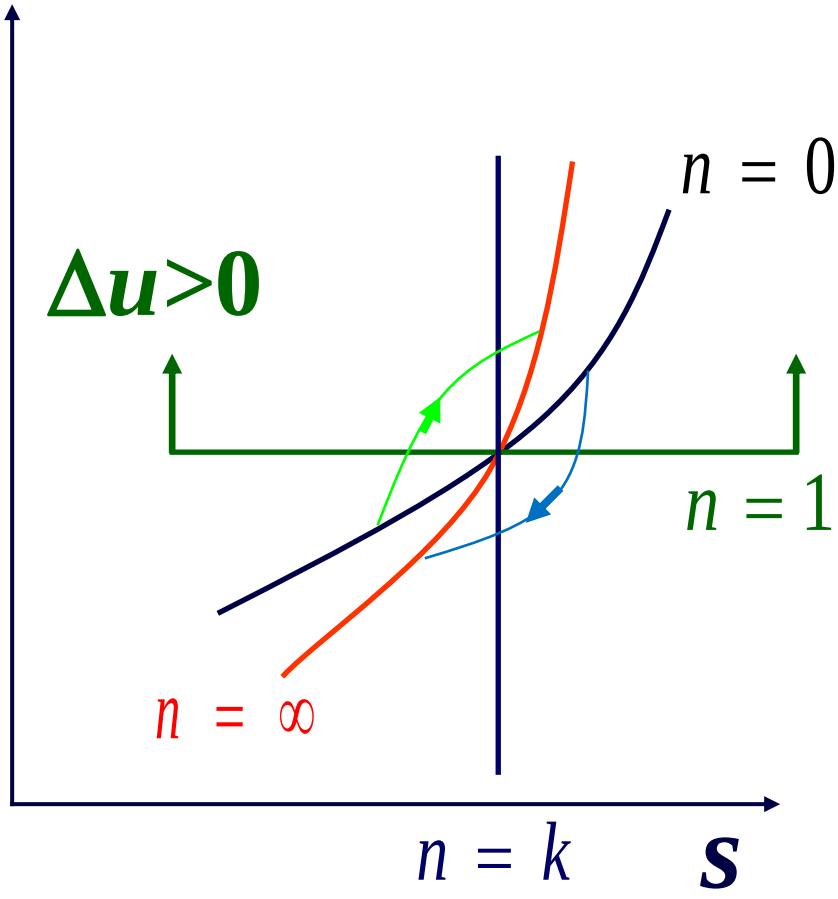




u在p-v,T-s图上的变化趋势

u = T ↑ $\Delta u = \int c_v dT$ ↑ $p_2 > p_1 \Rightarrow T_2 > T_1$



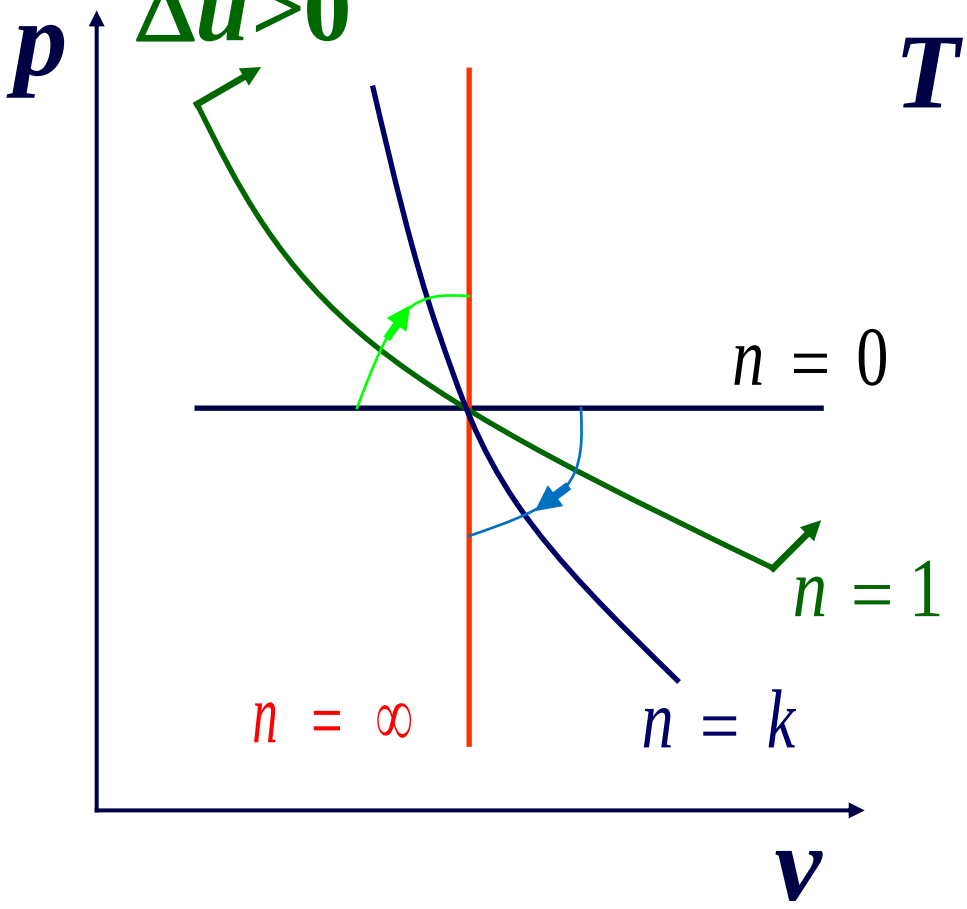


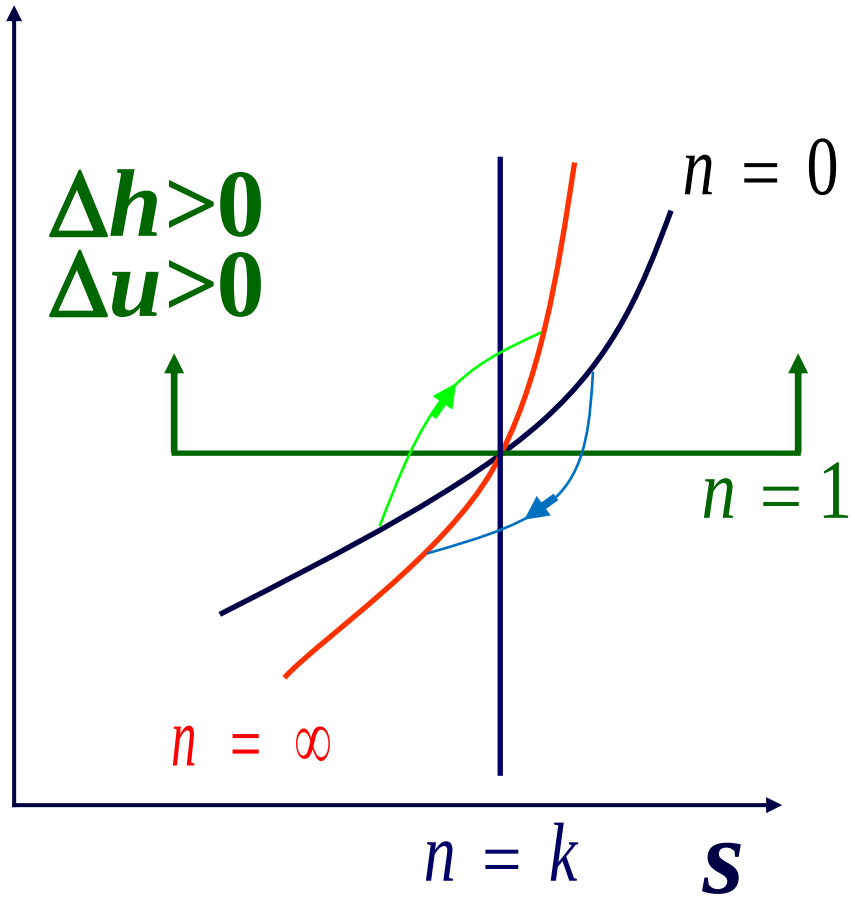
*h*在*p-v*,*T-s*图上的变化趋势

$h = T$

$\uparrow \Delta h = \int c_p dT \uparrow$

Δ*h*>0
Δ*u*>0

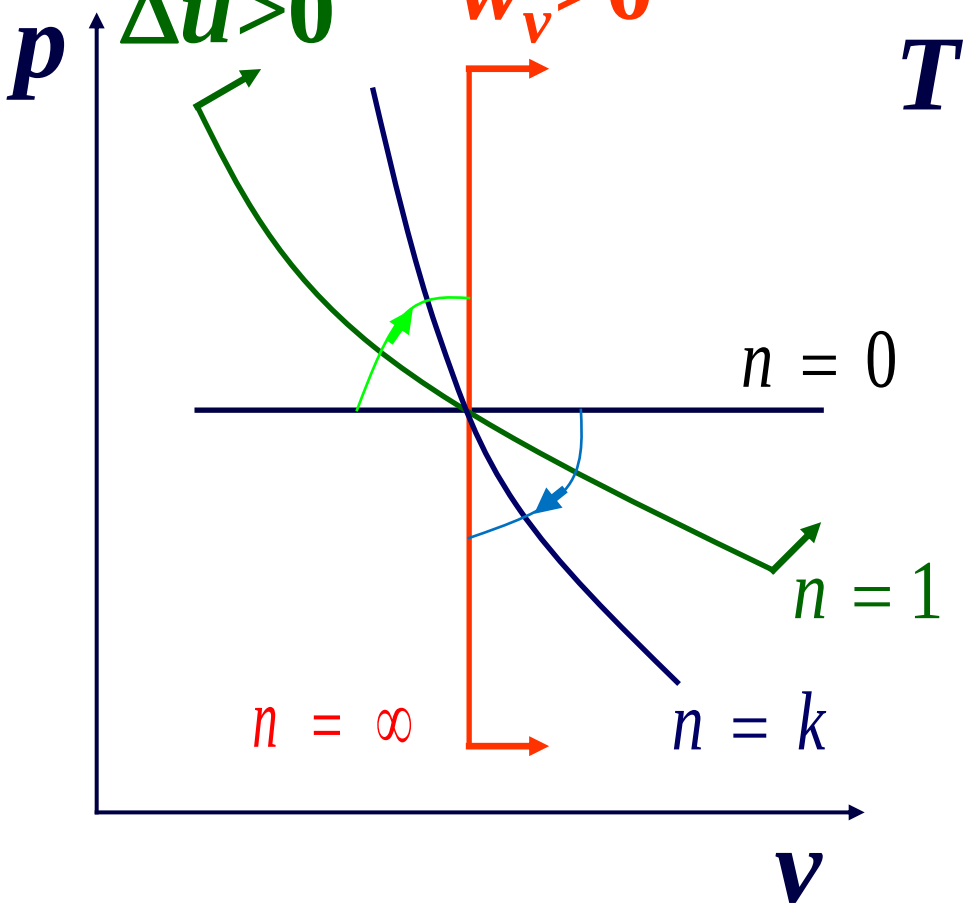


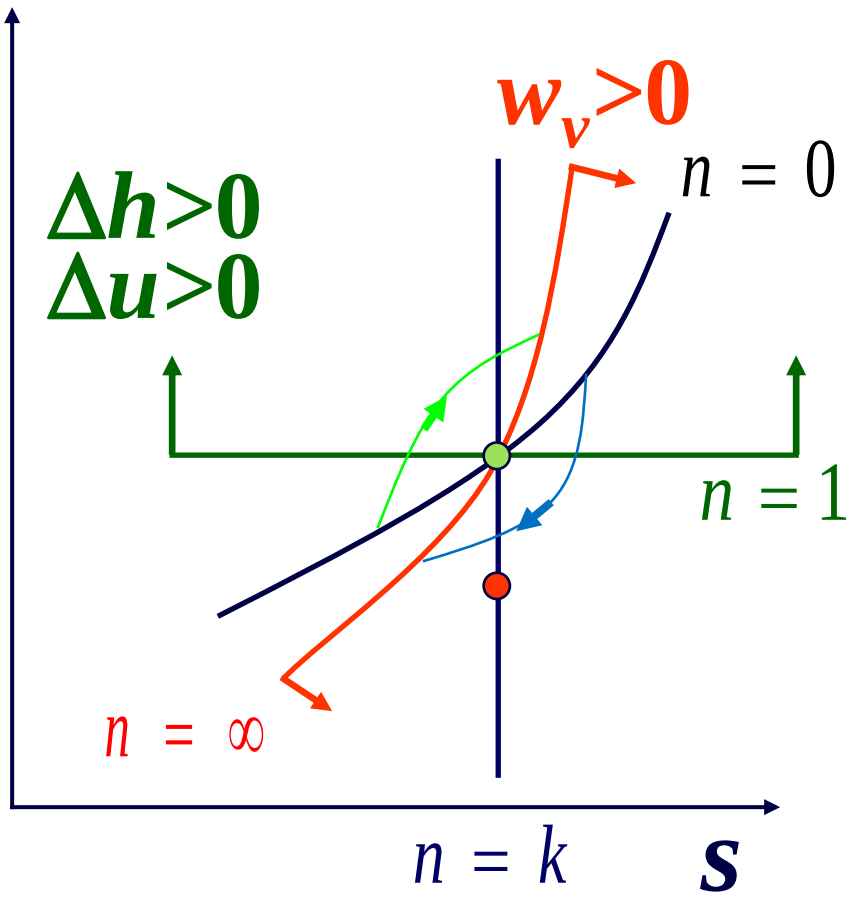


w_v在p-v,T-s图上的变化趋势

$$w = \int p\,dv$$

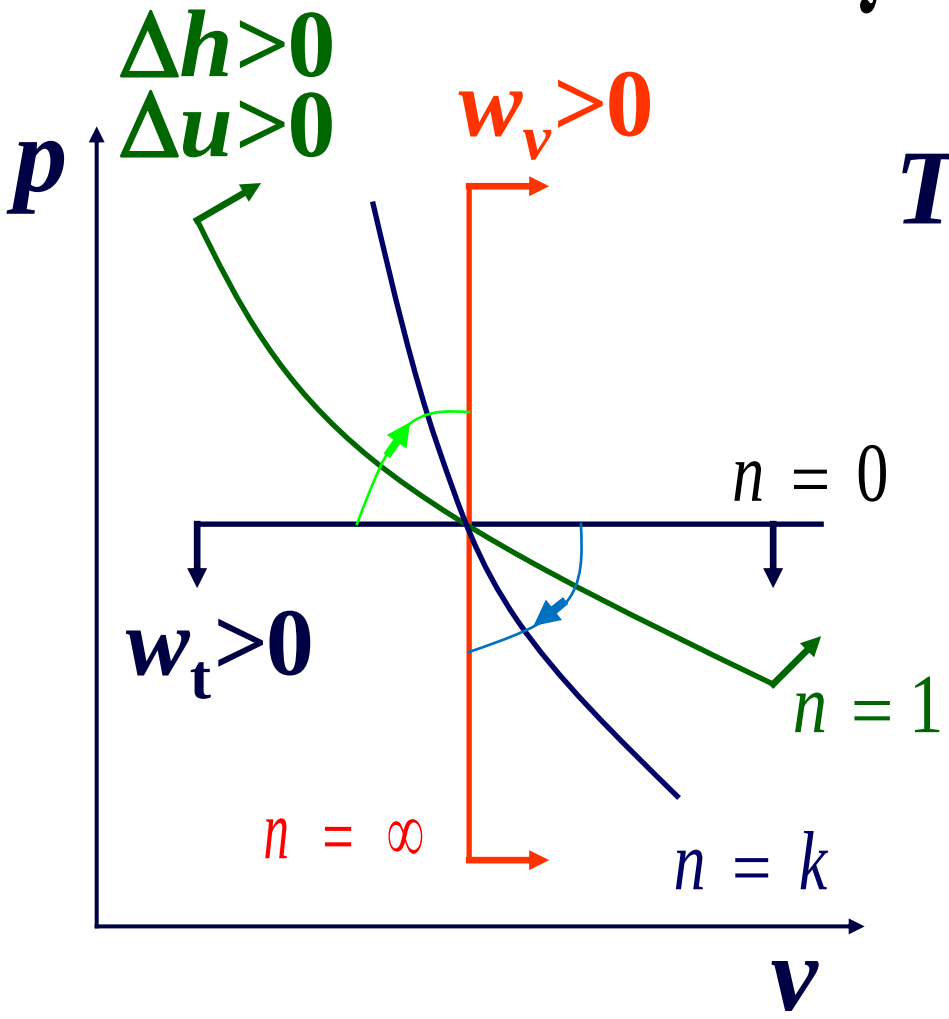
$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{k-1}$$

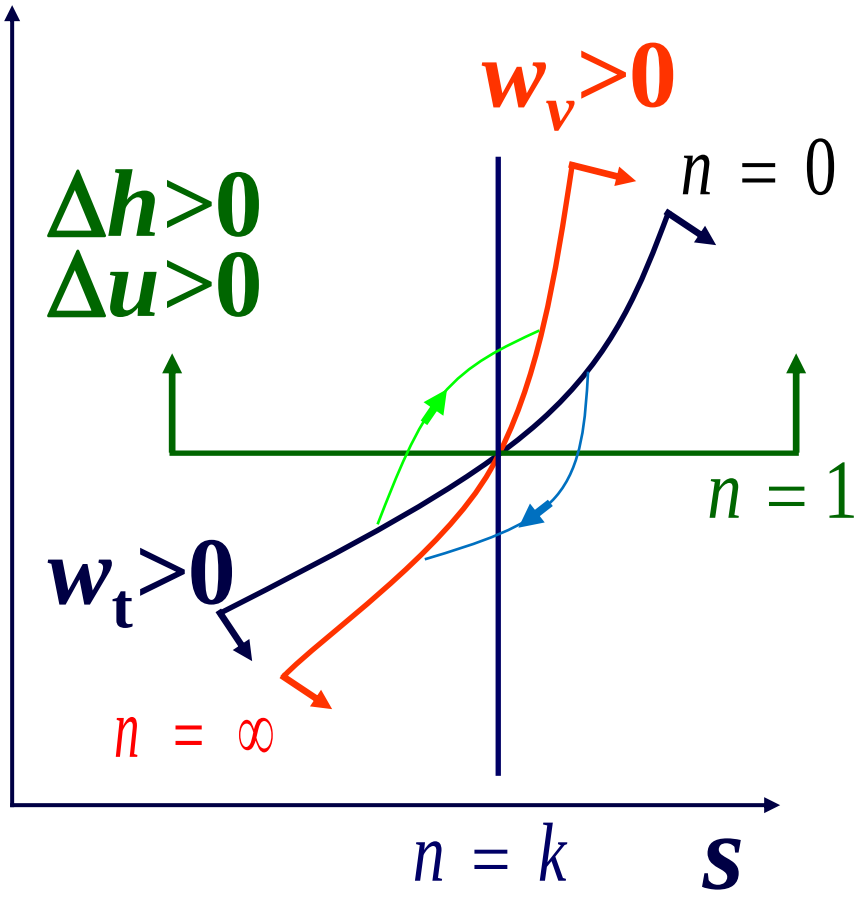




w_t在p-v,T-s图上的变化趋势

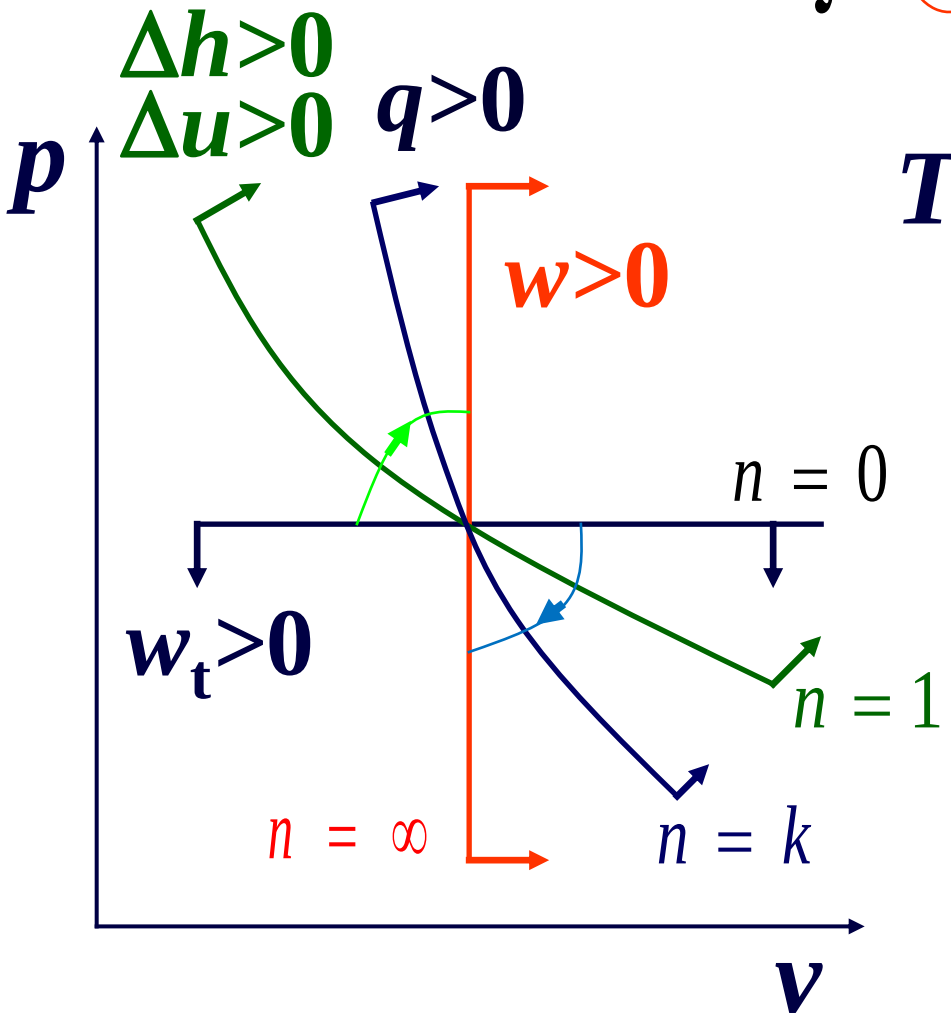
$$\uparrow w_t = -\int v\,dp \downarrow$$

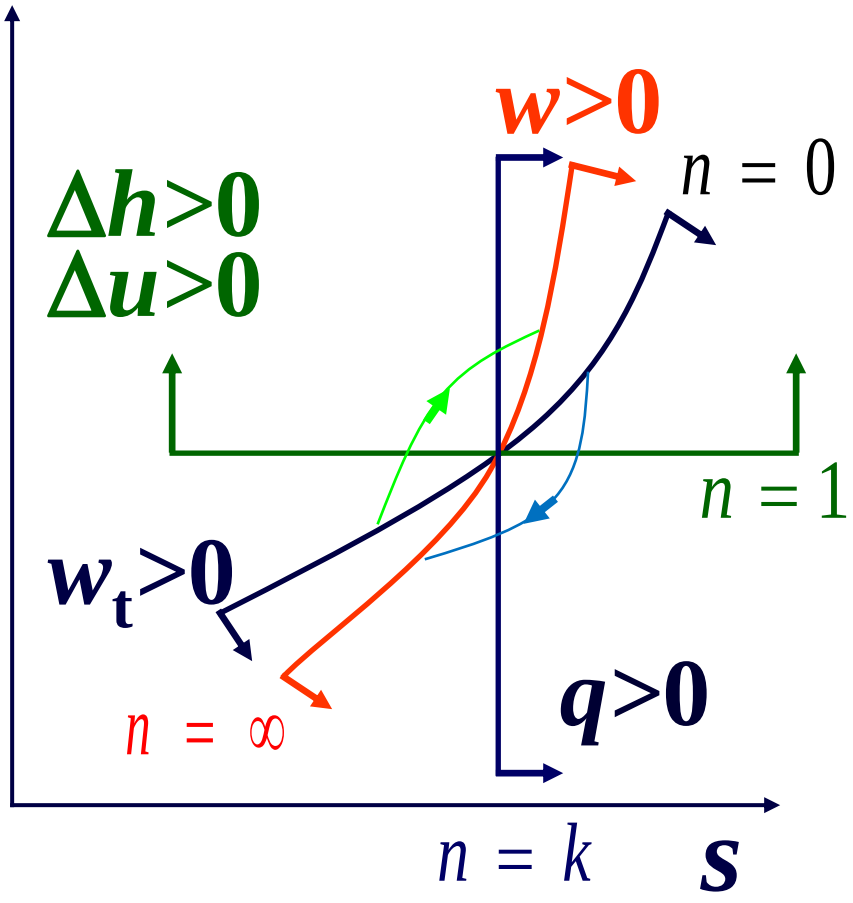




*q*在*p-v*,*T-s*图上的变化趋势

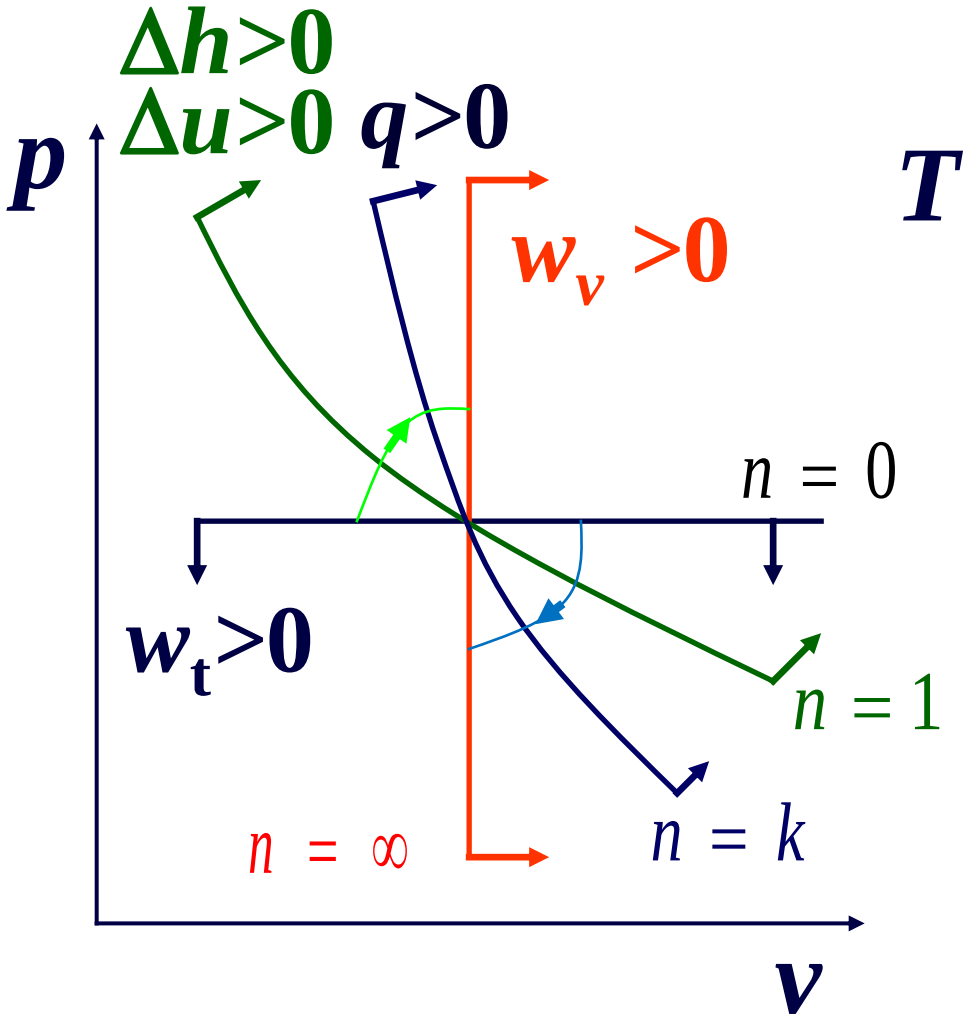
$$\uparrow q = \int T ds \uparrow$$

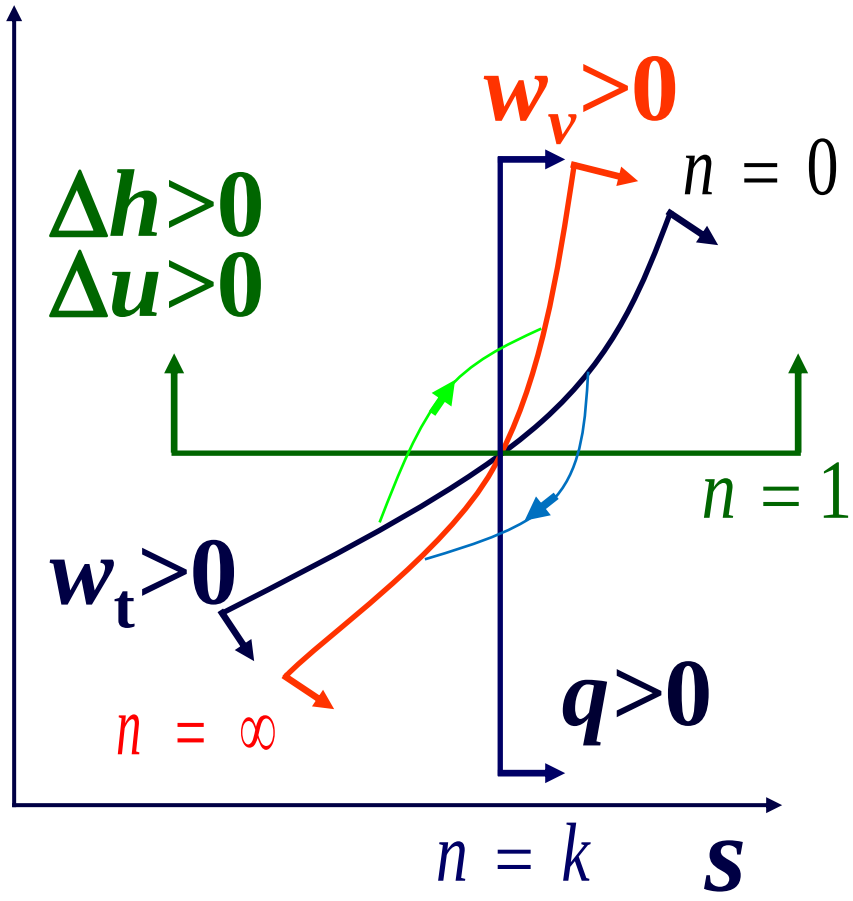




u, h, w_v, w_t, q变化趋势小结

$u\uparrow$, $h\uparrow(T\uparrow)$ $w_v\uparrow(v\uparrow)$ $w_t\uparrow(p\downarrow)$ $q\uparrow(s\uparrow)$





§8-4 热力过程的热二律分析举例

热量不仅有“量”的大小

还有“质”的高低

热一律揭示了能量传递和转换在“量”上的守恒关系

热二律指出了在能量的传递和转换过程中不可逆将导致能量在“质”上的贬值

*Ex*平衡方程

$$ex_{h1} + ex_q = ex_{h2} + w_t + w_l$$

其中 $$ex_q = \left(1 - \frac{T_0}{T}\right) \times q$$

$$ex_h = \left(h_1 - T_0 s_1\right) - \left(h_0 - T_0 s_0\right)$$

$$w_l = w_{max} - w = T_0 \Delta s_g$$

1. 可逆定温过程

$w_v = w_t = q$ 是否违反热力学第二定律？

$ex_{h1} + ex_q = ex_{h2} + w_t + w_l \qquad w_l = 0$

$w_t = ex_q + (ex_{h1} - ex_{h2})$ 其中：$ex_q = \left(1 - \frac{T_0}{T}\right) q$

又，根据*Ex*的定义有：

$$ex_{h1} - ex_{h2} = h_1 - h_2 - T_0(s_1 - s_2) = 0 - T_0(-\frac{q}{T}) = \frac{T_0}{T} q$$

故 $w_t = \left(1 - \frac{T_0}{T}\right) q + \frac{T_0}{T} q = q$

转变为功的能力来自两部分：热量*Ex*和工质*Ex*降

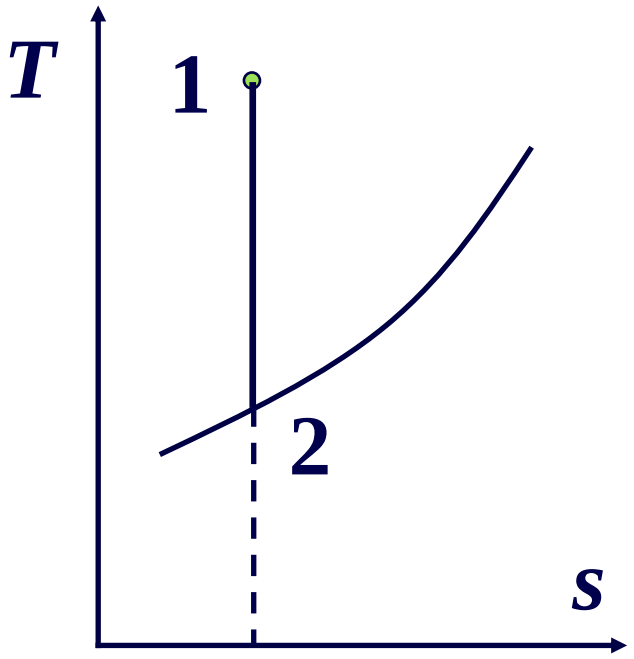
2. 绝热过程

1）可逆绝热过程

$$ex_{h1} + ex_q = ex_{h2} + w_t + w_l \qquad w_l = 0 \qquad ex_q = 0$$

$$ex_{h1} - ex_{h2} = w_t$$

工质*Ex*的减少量等于
其对外输出的技术功



根据*Ex*的定义：

$$ex_{h1} - ex_{h2} = h_1 - h_2 - T_0(s_1 - s_2)$$
$$= h_1 - h_2$$

膨胀*Ex*降对外作功 压缩 输入功变为工质*Ex*增

2. 绝热过程

2）不可逆绝热过程

$$ex_{h1} + ex_q = ex_{h2} + w_t + w_l \qquad ex_q = 0$$

$$ex_{h1} - ex_{h2} = w_t + w_l$$

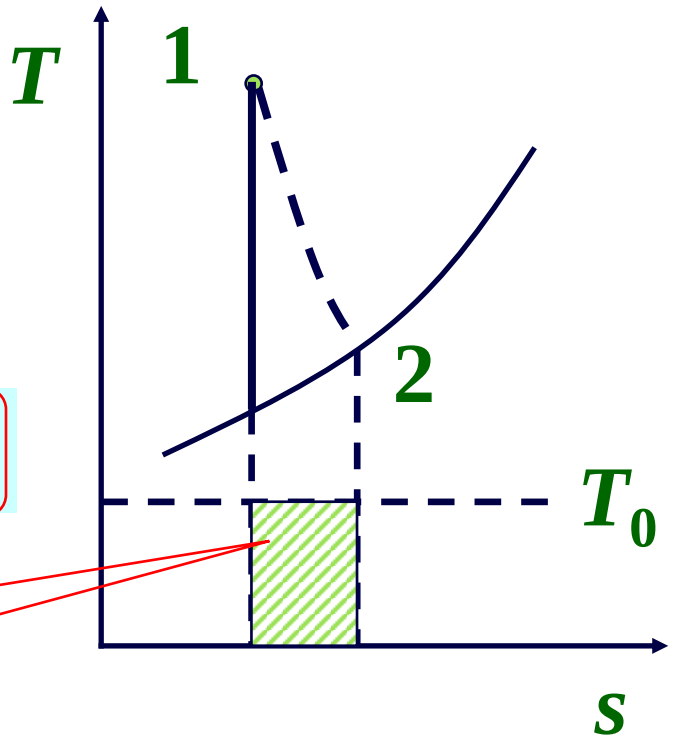
***Ex*降等于对外输出功与有效能损失之和**

根据*Ex*的定义：

$$ex_{h1} - ex_{h2} = h_1 - h_2 + T_0(s_2 - s_1)$$

$h_1 - h_2$ → w_t

$T_0(s_2 - s_1)$ → ***Ex*损失**



第八章 完

End of Chapter Eight

第八章 小结 Summary

- 理想气体四个基本过程的特性，参数变化，功和热的计算
- 理想气体多变过程的特性，参数变化，功和热的计算
  基本过程是多变过程的特例
- 各种过程在 *p*-*v*图和*T*-*s*图上的表示
- 热力过程的热二律分析

